

# Mahmoud Souabni

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Portfolio: <https://mahmon5.github.io/Portfolio/>

Engineering Student in Industrial Computing & Automation – Final Year (Engineering Degree)

## Profile

Final-year engineering student (M.Eng., Final Year) at INSAT, specializing in embedded systems, mobile robotics, and industrial automation. Completed a recent 2-month internship at CyberMech Systems configuring autonomous navigation (ROS 2, Nav2, SLAM) for a modular mobile robot. Additional experience in microcontrollers (STM32, ESP32, NXP), PCB design, and CAD. Seeking a final-year (capstone) internship for 2027 in robotics, embedded systems, or industrial automation.

## Education

2024–2027 **Engineering Degree (Industrial Computing & Automation) – 3<sup>rd</sup> Year (in progress)**, INSAT, Tunis

Modules: Control Systems, Automation, C/C++, Microprocessor Architecture, Databases.

2022–2024 **Preparatory Cycle (Mathematics, Physics, Computer Science)**, INSAT, Tunis

## Professional Experience

Summer 2026 **Intern – Autonomous Navigation, AMR-X Robot, CyberMech Systems**, Tunis

(2 months) Configured the ROS 2/Nav2 navigation stack for a modular mobile robot (AMR-X) in Gazebo: SLAM mapping, AMCL localization, path planning, obstacle avoidance, keepout zones, and mission execution via named stations. Diagnosed and corrected an odometry drift issue (kinematic calibration). Contributed to the CAD → URDF pipeline and to a market study of the AMR sector.

Summer 2025 **Observation Internship, Tunisair Technics**, Tunis

MRO: hands-on exposure to aircraft maintenance, avionics, and quality control.

## Technical Skills

|                    |                                                                                   |
|--------------------|-----------------------------------------------------------------------------------|
| <b>Programming</b> | C/C++, Python, MATLAB, Java, SQL, Assembly                                        |
| <b>Robotics</b>    | ROS 2 (Jazzy)   Nav2, SLAM Toolbox, AMCL   Gazebo Harmonic, PyBullet   Git/colcon |
| <b>Embedded</b>    | Arduino, ESP32, STM32, NXP   RTOS   Protocols (I2C, SPI, UART)                    |
| <b>Automation</b>  | TIA Portal, Factory I/O, PL7 Pro, MATLAB/Simulink                                 |
| <b>Design</b>      | SolidWorks, URDF   EasyEDA, Proteus (PCB Design and Simulation)                   |

## Technical Projects

2025–2026 **Spider Robot – Final-Year Project (INSAT)** – Quadruped robot: 3D modeling, custom PCB, C firmware (servo control), and gait simulation in PyBullet.

2024 **AI-Based Beamforming (MATLAB)** – LSTM neural network for optimizing antenna beam direction.

## Competitions & Robotics

2026 **NXP Cup Tunisia** – Programmed an autonomous car on an NXP microcontroller.

2025 **ECPP 2.0 (IEEE RAS INSAT)** – Competitive programming contest focused on embedded systems.

2024–2025 **Line Follower Robot** – PCB and STM32/ESP32 firmware (Robolympix, Cyberbot, ENSI Robocup).

## Activities & Leadership

2024–Pr. **Aerobotix INSAT** – Active member of the competitive robotics team.

2022–2024 **IEEE Student Branch (CS/RAS) & Securinets INSAT**.

## Languages

**Arabic:** Native | **French:** Professional | **English:** Professional